2.2 Handshake signaling

This section gives details of the handshake signaling and defines the relationship of the **TVALID** and **TREADY** signals.

The **TVALID** and **TREADY** handshake determines when information is passed across the interface. A two-way flow control mechanism enables both the Transmitter and Receiver to control the rate at which the data and control information is transmitted across the interface.

For a transfer to occur, both **TVALID** and **TREADY** must be asserted. Either **TVALID** or **TREADY** can be asserted first, or both can be asserted in the same **ACLK** cycle.

A Transmitter is not permitted to wait until **TREADY** is asserted before asserting **TVALID**. Once **TVALID** is asserted, it must remain asserted until the handshake occurs.

A Receiver is permitted to wait for **TVALID** to be asserted before asserting **TREADY**. It is permitted that a Receiver asserts and deasserts **TREADY** without **TVALID** being asserted.

The following sections give examples of the handshake sequence.

2.2.1 Handshake with TVALID asserted before TREADY

In Figure 2-1, the Transmitter presents the data and control information and asserts **TVALID** as HIGH. The valid data bytes and control information from the Transmitter must remain unchanged once **TVALID** has been asserted by the Transmitter. The Receiver can accept the data and control information once **TREADY** is HIGH. The transfer takes place once the Receiver asserts **TREADY** HIGH. Figure 2-1 shows the transfer occurs at T3.

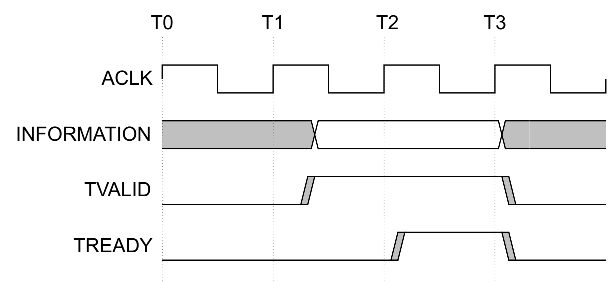

Figure 2-1 Handshake with TVALID asserted before TREADY

2.2.2 Handshake with TREADY asserted before TVALID

In Figure 2-2, the Receiver drives **TREADY** HIGH before the data and control information is valid. This indicates the Receiver can accept the data and control information in a single **ACLK** cycle. In this case, the transfer occurs once the Transmitter drives **TVALID** HIGH. Figure 2-2 shows the transfer occurring at T3.

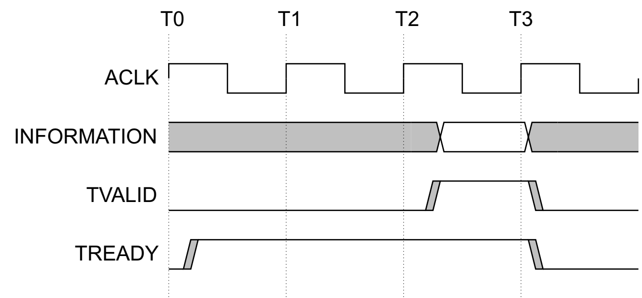

Figure 2-2 Handshake with TREADY asserted before TVALID

2.2.3 Handshake with TVALID and TREADY asserted simultaneously

In Figure 2-3, the Transmitter asserts **TVALID** HIGH and the Receiver asserts **TREADY** HIGH in the same **ACLK** cycle. In this case, transfer takes place in the same cycle, as shown in T2 of Figure 2-3.

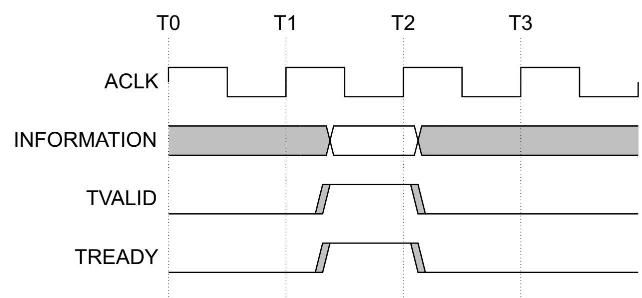

Figure 2-3 Handshake with TVALID and TREADY asserted simultaneously